UNIT 4 · TOPIC 4.3

Protecting Devices

Assignment

Instructions: 17 practice questions + 3 coding assignment(s), 32 points total. Show your work for short-answer items. Coding assignments may be completed here or in the interactive auto-graded environment on the course site (same problem, same points).

Name: **Date:** **Class:**

Practice Questions

4.3.a · Device hardening

1. A network administrator sets up a new router and immediately connects it to the internet without changing any settings. What is the PRIMARY security risk? (1 pt)

- ○ A. The router firmware is too new to be stable
- ○ B. The router will use too much bandwidth
- ○ C. The router cannot route traffic until configured
- ○ D. Default credentials allow any attacker to log in as administrator

2. Which action BEST describes the principle of least privilege when hardening a workstation? (1 pt)

- ○ A. Giving every user administrator rights so they can install updates
- ○ B. Disabling all accounts except the root account
- ○ C. Sharing a single administrator account among all staff to simplify management
- ○ D. Granting each user account only the permissions needed for their job function

3. Select ALL items that are part of device hardening. (Choose all that apply.) (3 pts)

- □ A. Change default passwords
- □ B. Disable services not needed for the device's function
- □ C. Enable full-disk encryption
- □ D. Enable Telnet for easy remote access

(Select ALL that apply.)

4. What is 'configuration drift' in the context of device hardening? (1 pt)

- ○ A. When a device physically moves to a new network location
- ○ B. The gradual accumulation of unauthorized or unintended configuration changes over time
- ○ C. The planned update of firmware to a new version
- ○ D. The process of migrating from one security baseline to another

5. Name the term for the total set of ways an attacker could interact with or exploit a device. (1 pt)

4.3.b · Antivirus and endpoint protection

6. An administrator wants to detect a brand-new, customized piece of malware that does not match any known virus definition database. Which detection method is MOST likely to identify this threat? (1 pt)

- ○ A. Signature-based detection
- ○ B. Heuristic or behavioral analysis
- ○ C. Hashing-based verification
- ○ D. Passive packet sniffing

7. What is a primary drawback of relying heavily on behavioral analysis for malware detection? (1 pt)

- ○ A. It cannot detect zero-day exploits.
- ○ B. It requires a constantly updated signature database to function.
- ○ C. It only operates when the computer is disconnected from the network.
- ○ D. It may generate a higher rate of false positives on legitimate administrative software.

8. Select ALL functions that a host-based firewall typically performs. (Choose all that apply.) (3 pts)

- □ A. Blocking unauthorized inbound connections to local services
- □ B. Preventing unauthorized outbound connections from local malware to command servers
- □ C. Scanning local files for known virus signatures
- □ D. Filtering traffic specifically for the device on which it is installed

(Select ALL that apply.)

9. What is the term for a security alert that incorrectly flags a harmless file or action as malicious? (1 pt)

10. An attacker abuses PowerShell, a legitimate pre-installed Windows tool, to download and run malicious code instead of using custom malware. What is this technique called? (1 pt)

- ○ A. Living-off-the-land
- ○ B. Signature spoofing
- ○ C. Polymorphic encoding
- ○ D. Heuristic bypass

11. How does Endpoint Detection and Response (EDR) primarily differ from traditional antivirus? (1 pt)

- ○ A. EDR only works on servers, never on laptops or desktops
- ○ B. EDR continuously records endpoint activity and gives responders tools to investigate and contain threats retroactively, beyond just blocking at scan time
- ○ C. EDR replaces the need for any signature-based scanning
- ○ D. EDR requires no central console and runs independently on each device

4.3.c · Patching and regular updates

12. What is a 'zero-day' vulnerability? (1 pt)

- ○ A. A vulnerability that has been patched for zero days, meaning it was just fixed
- ○ B. A vulnerability with a CVSS score of exactly zero
- ○ C. A vulnerability being actively exploited before any patch exists

○ D. A vulnerability that only affects devices zero days old

13. Most real-world breaches exploit which type of vulnerability? (1 pt)

○ A. Zero-days, because no patch can stop them

○ B. N-days, because a patch exists but the target has not yet applied it

○ C. Vulnerabilities with a CVSS score below 1.0

○ D. Only vulnerabilities in end-of-life software

14. Select ALL legitimate reasons an organization might delay applying a patch immediately. (Choose all that apply.) (3 pts)

□ A. The patch has not yet been tested and could break a business-critical application

□ B. Applying the patch requires a scheduled reboot that must avoid business hours

□ C. The vulnerability does not actually exist and was invented by the vendor

□ D. The organization has thousands of endpoints requiring a staged rollout

(Select ALL that apply.)

15. A public-facing web server has a critical CVSS 9.1 vulnerability that CISA confirms is being actively exploited in the wild right now. What should happen? (1 pt)

○ A. Apply emergency out-of-cycle patching as urgently as possible

○ B. Ignore it since the server has worked fine so far

○ C. Schedule the patch for the next routine monthly maintenance window

○ D. Wait for an end user to report a problem before acting

16. Name the term for software that no longer receives vendor security patches, requiring compensating controls instead of patching. (1 pt)

17. Why does publishing a patch increase urgency to apply it quickly? (1 pt)

○ A. Patches always contain new features attackers want to steal

○ B. Patches disable the antivirus on the system automatically

○ C. The patch itself (or its public disclosure) reveals details attackers can reverse-engineer into a working exploit

○ D. Publishing a patch has no effect on attacker behavior

Coding Assignments (Python 3)

4.3.a · Device hardening

3 pts · auto-graded on the site

Coding problem, 4.3.a: Device hardening configuration auditor

Skill: Mitigate Risk · **Maps to:** 4.3.A · **Language:** Python 3 (pytest)

Problem

In a secure enterprise environment, devices must be configured according to strict security baselines before deployment. Over time, administrators or software installers may change settings, resulting in "configuration drift" that increases the device's attack surface.

Your task is to write a python function `solve(data)` that takes a target device's actual configuration and audit it against a security baseline. The function must return a list of all check names that the configuration fails, sorted alphabetically.

Function contract

Implement `solve(data)` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

A dictionary with keys `"baseline"` and `"configuration"`. Example:

```
{
  "baseline": {
    "min_password_length": 12,
    "disallowed_services": ["ftp", "telnet"],
    "firewall_enabled": true,
    "encryption_enabled": true,
    "allowed_accounts": ["admin", "operator"]
  },
  "configuration": {
    "password_length": 8,
    "services": ["ssh", "ftp"],
    "firewall_enabled": false,
    "encryption_enabled": true,
    "accounts": ["admin", "guest", "operator"]
  }
}
```

Output

A list of strings representing the names of the failing checks from the baseline, sorted alphabetically. Example output for the input above:

`["allowed_accounts", "disallowed_services", "firewall_enabled", "min_password_length"]`

(Explanation: password length 8 is less than 12; ftp service is disallowed; firewall is false; guest account is not allowed)

Constraints

- If an expected configuration field is missing, default values apply: - `password_length`: `0` - `services`: `[]` - `firewall_enabled`: `False` - `encryption_enabled`: `False` - `accounts`: `[]`
- Standard library only.
- Case-sensitive string matching for services and accounts.

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
def solve(data: dict) -> list[str]:
    """Audit a device's configuration against a security baseline.

    Args:
        data: A dict with keys "baseline" and "configuration".

    Returns:
        A list of failing baseline check names, sorted alphabetically.
    """
    raise NotImplementedError("Implement this function")
```

4.3.b · Antivirus and endpoint protection

3 pts · auto-graded on the site

Coding problem, 4.3.b: Endpoint Detection Engine

Skill: Mitigate Risk · **Maps to:** 4.3.B · **Language:** Python 3 (pytest)

Problem

Endpoint Protection Platforms (EPP) use a mixture of signature detection (known hashes) and behavioral heuristics (actions taken by the process, such as excessive file updates or network socket connections) to identify and suspend threats.

You are tasked with writing a process-auditing rule engine `solve(data)`. The engine evaluates a list of running processes against: 1. A database of known malicious file hashes (signatures). 2. Max thresholds on process behavior: `max_file_modifications` and `max_network_connections`.

The function must return a list of PIDs of all processes that violate any of these rules.

Function contract

Implement `solve(data)` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

A dictionary containing security signatures, behavioral thresholds, and a list of active processes:

```
{
  "signatures": ["d41d8cd98f00b204e9800998ecf8427e", "098f6bcd4621d373cade4e832627b4f6"],
  "behavioral_rules": {
    "max_file_modifications": 10,
    "max_network_connections": 5
  },
  "processes": [
    {
      "pid": 101,
      "name": "backup_agent.exe",
      "hash": "81dc9bdb52d04dc20036dbd8313ed055",
      "file_modifications": 8,
      "network_connections": 2
    },
    {
      "pid": 102,
      "name": "ransomware.exe",
      "hash": "d41d8cd98f00b204e9800998ecf8427e",
      "file_modifications": 2,
      "network_connections": 0
    },
    {
      "pid": 103,
      "name": "spam_sender.exe",
      "hash": "1f10c9aa0090ff91e90ff8a8bc8a9bcf",
      "file_modifications": 1,
      "network_connections": 12
    }
  ]
}
```

Output

A list of integers representing the PIDs of flagged processes, sorted in ascending order. For the example input above:

```
[102, 103]
```

(Explanation: PID 102 matches a known signature; PID 103 exceeds the network connections threshold)

Constraints

- Signature matching is case-insensitive (e.g., `"A1B2"` matches `"a1b2"`).
- Behavioral violations occur when a metric is *strictly greater* than the threshold (i.e. `file_modifications > max_file_modifications`).
- Standard library only.

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 4.3.b. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""


def solve(data: dict) -> list:
    """Flag processes that match a known-malware signature or exceed behavioral thresholds.

    Args:
        data: dict with keys "signatures" (list[str]), "behavioral_rules" (dict with
            "max_file_modifications" and "max_network_connections"), and "processes"
            (list of dicts with pid, name, hash, file_modifications, network_connections).

    Returns:
        Sorted list of PIDs (int) for every process that matches a signature or
        exceeds a behavioral threshold.
    """
    # TODO: implement per prompt.md
    raise NotImplementedError("implement solve() — see prompt.md")
```

4.3.c · Patching and regular updates

3 pts · auto-graded on the site

Coding problem, 4.3.c: Patch-prioritization engine

Skill: Mitigate Risk · **Maps to:** 4.3.B, 4.3.C · **Language:** Python 3 (pytest)

Problem

A patch-management team needs to triage a queue of pending vulnerabilities and decide how urgently each must be addressed. Each vulnerability has a CVSS severity score, a flag for whether it is being actively exploited in the wild, a flag for whether the affected asset is internet-facing, and a flag for whether the affected software is end-of-life (no patch will ever be released). Build the triage engine that assigns one urgency label to each vulnerability.

Function contract

Implement `def solve(vulns: list) -> list` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

A list of dicts, each with keys: `id` (str), `cvss` (float, 0.0-10.0), `exploited_in_wild` (bool), `internet_facing` (bool), `end_of_life` (bool).

Output

A list of dicts, one per input vulnerability (same order as input), each with keys `id` (str, copied from input) and `urgency` (str), where `urgency` is one of: `"emergency"`, `"scheduled"`, `"routine"`, `"compensate"`.

Decision rule (apply in this exact order, first match wins, per vulnerability)

1. `end_of_life` is `True` → `"compensate"` (no patch will ever exist; patching is not an option regardless of severity or exploitation status). 2. Else, `exploited_in_wild` is `True` AND `internet_facing` is `True` → `"emergency"`. 3. Else, `cvss >= 9.0` AND `internet_facing` is `True` → `"emergency"` (a working exploit is likely imminent for a critical, internet-facing flaw even without confirmed in-the-wild activity). 4. Else, `cvss >= 9.0` → `"scheduled"` (critical severity but not internet-facing). 5. Else, `cvss >= 7.0` AND `internet_facing` is `True` → `"scheduled"`. 6. Else → `"routine"`.

Rule order matters: `end_of_life` is checked first and overrides every other condition, an end-of-life system is never `"emergency"` or `"scheduled"` even at CVSS 10.0, because patching is not a valid remediation at all.

Examples

```
[{"id": "V1", "cvss": 9.1, "exploited_in_wild": True, "internet_facing": True, "end_of_life": False}]
    -> [{"id": "V1", "urgency": "emergency"}]

[{"id": "V2", "cvss": 8.8, "exploited_in_wild": False, "internet_facing": False, "end_of_life": True}]
    -> [{"id": "V2", "urgency": "compensate"}]

[{"id": "V3", "cvss": 4.2, "exploited_in_wild": False, "internet_facing": False, "end_of_life": False}]
    -> [{"id": "V3", "urgency": "routine"}]
```

Constraints

- Deterministic, offline, standard library only.
- Preserve input order in the output list.
- Exactly one urgency label per vulnerability.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 4.3.c. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""


def solve(vulns: list) -> list:
    """Assign an urgency label to each vulnerability per the patch-triage priority rules.

    Args:
        vulns: list of dicts with keys id (str), cvss (float), exploited_in_wild (bool),
            internet_facing (bool), end_of_life (bool).

    Returns:
        list of dicts, same order, each {"id": ..., "urgency": ...} where urgency is one of
        "emergency", "scheduled", "routine", "compensate".
    """
    # TODO: implement per prompt.md
    raise NotImplementedError("implement solve() — see prompt.md")
```